นายปริญญ์นกร อยู่แท้กูล 673380277-1

Packet Tracer - TCP and UDP Communications

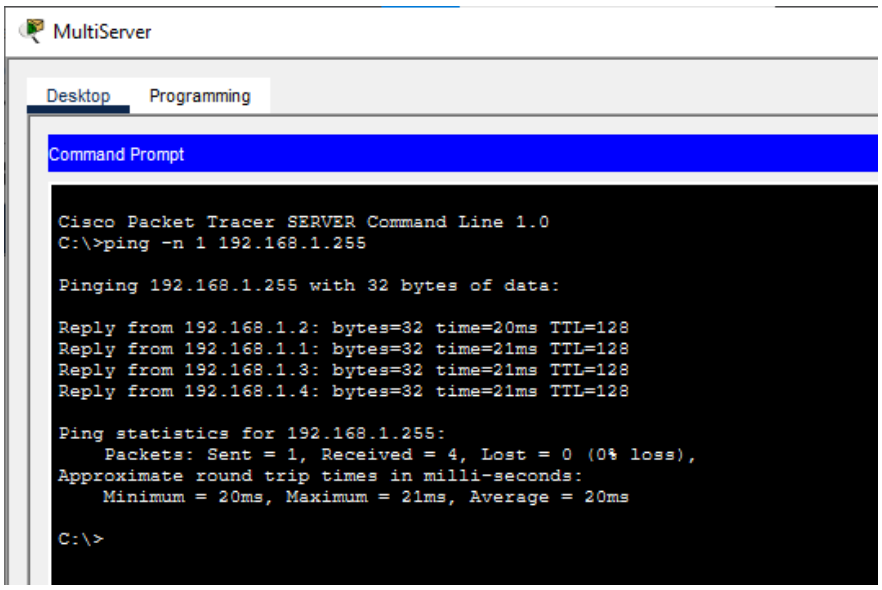
Background

This simulation activity is intended to provide a foundation for understanding TCP and UDP in detail. Packet Tracer simulation mode provides you the ability to view the state of different PDUs as they travel through the network.

Packet Tracer Simulation mode enables you to view each of the protocols and the associated PDUs. The steps outlined below lead you through the process of requesting network services using various applications that are available on a client PC. You will explore the functionality of the TCP and UDP protocols, multiplexing, and the function of port numbers in determining which local application requested the data or is sending the data. Packet Tracer will not score this activity.

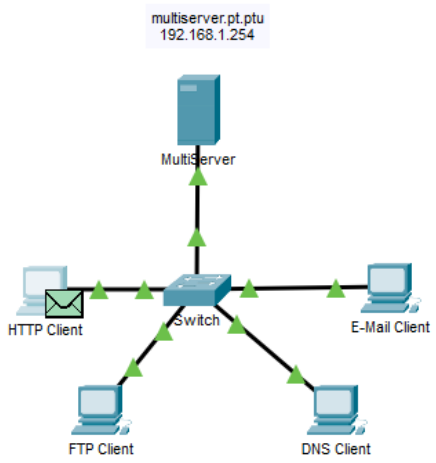
Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

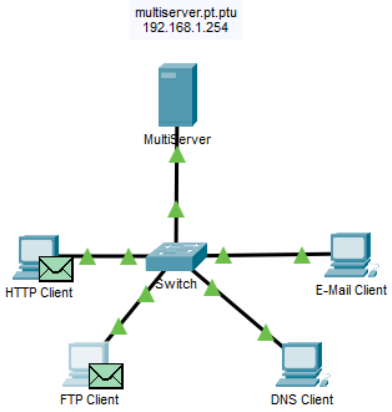


จะเห็นได้ว่าคำสั่งนี้จะมีการส่ง ping ไปยังทุกเครื่องในวง LAN แค่ครั้งเดียว และส่งไปสำเร็จทั้งหมด

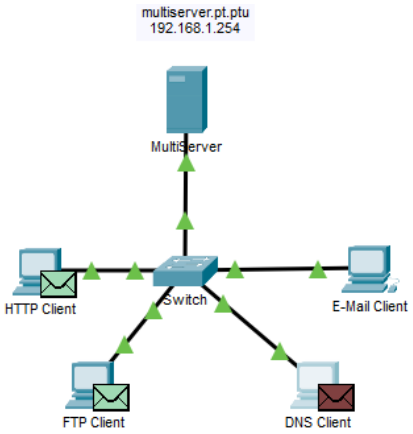
Step 2: Generate web (HTTP) traffic.



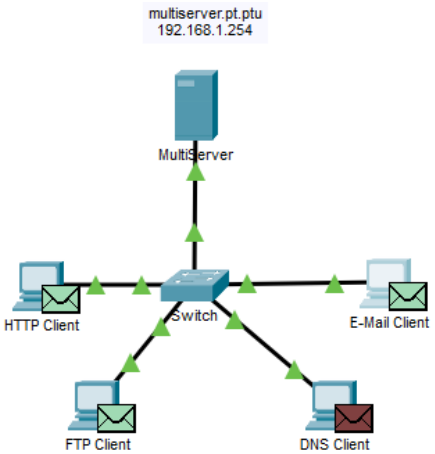
Step 3: Generate FTP traffic.



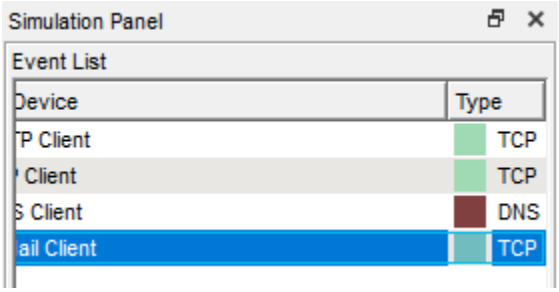
Step 4: Generate DNS traffic.



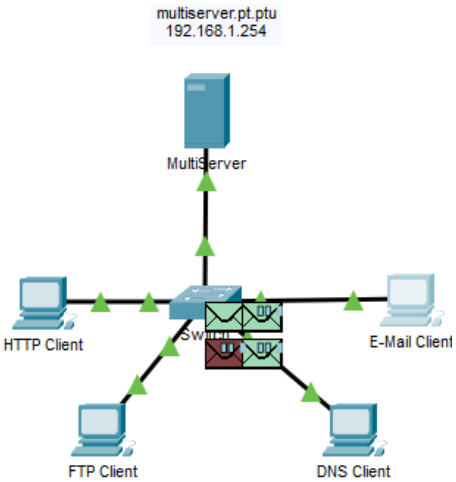
Step 5: Generate Email traffic.



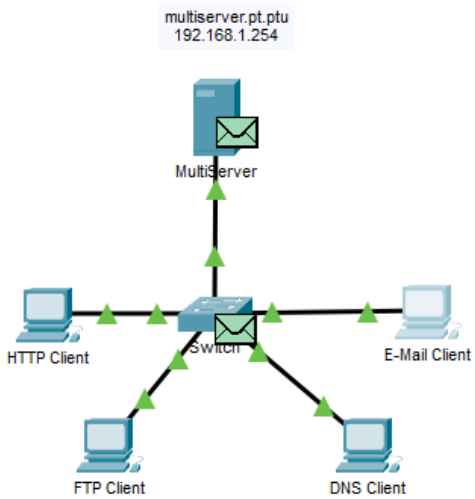
Step 6: Verify that the traffic is generated and ready for simulation.



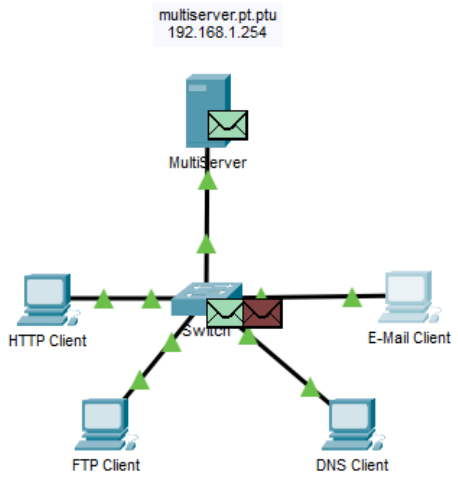
Step 7: Examine multiplexing as the traffic crosses the network.



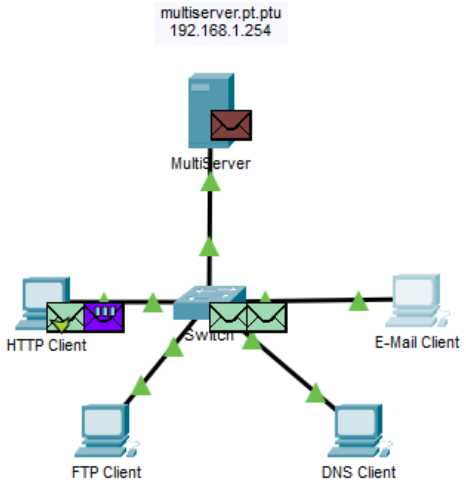
1 click



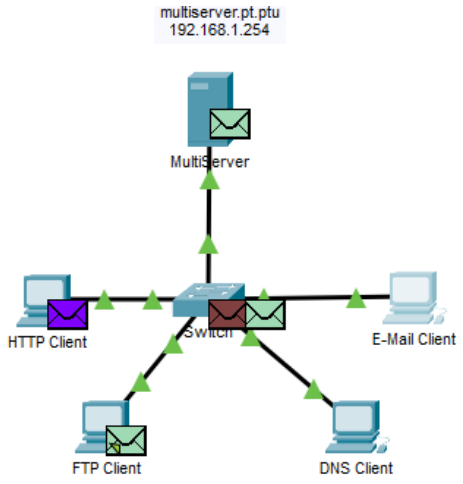
2 click



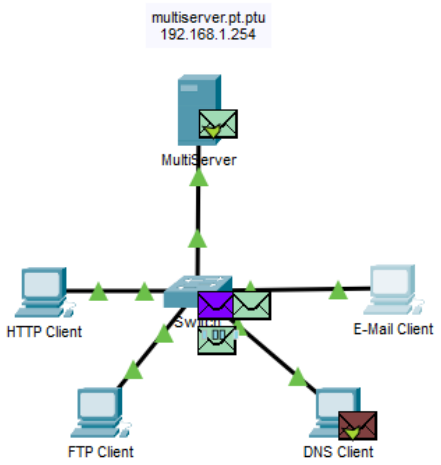
3 click



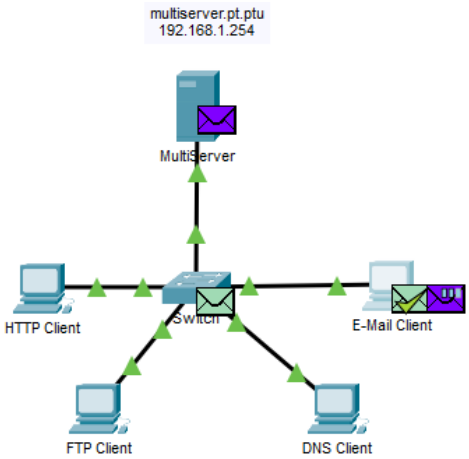
4 click



5 click



6 click



รวมคำถาม

What is this called?

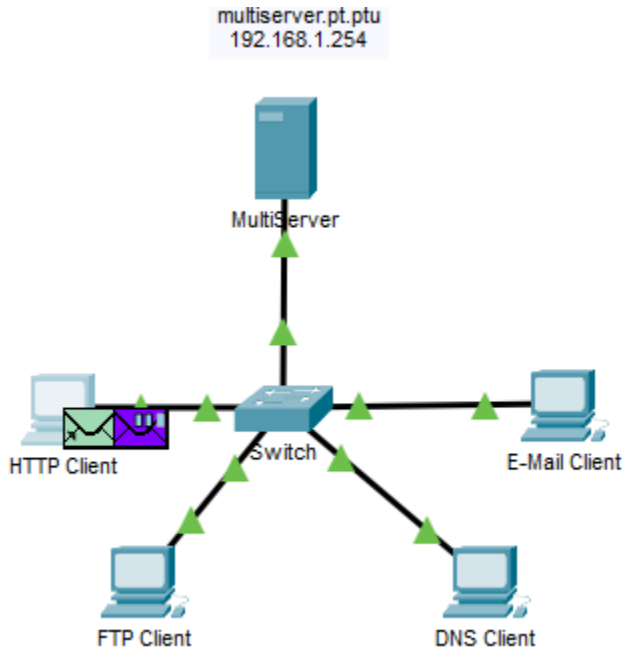
ตอบ เรียกว่า Multiplexing คือการสลับกันส่งข้อมูลไปมาในสายเดียวกัน

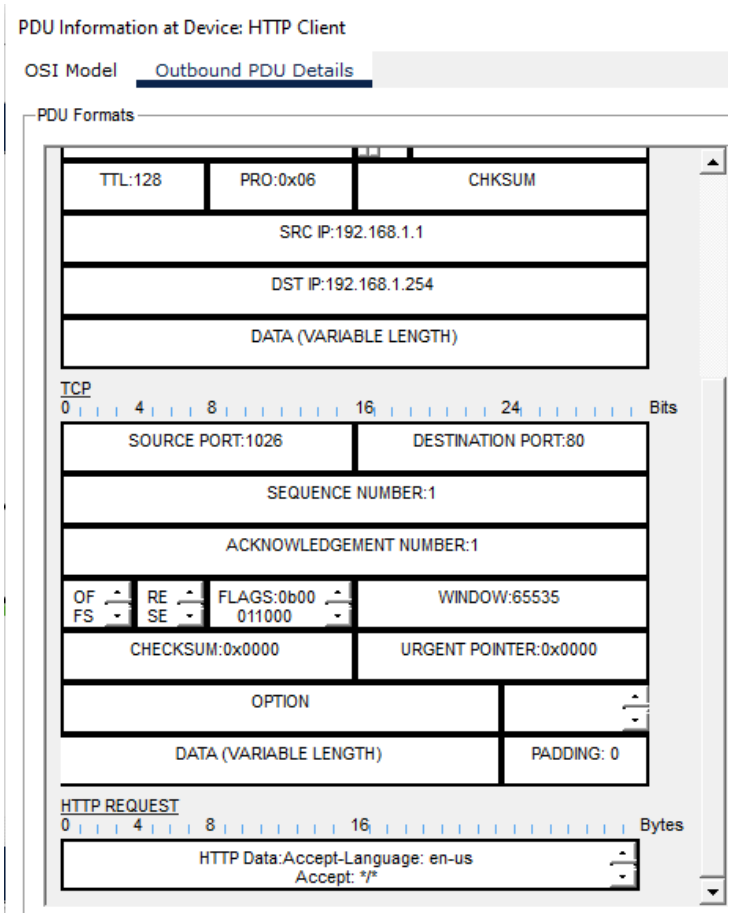
A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?

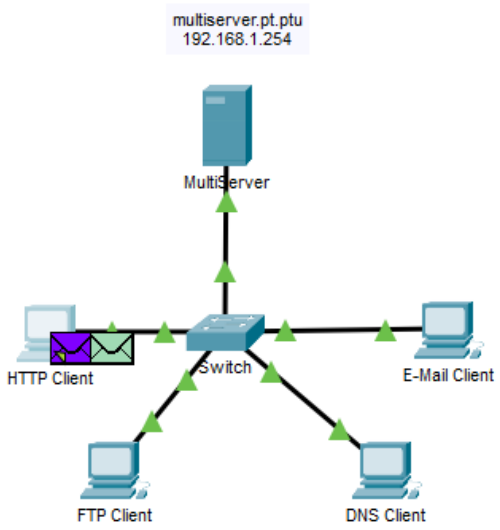
ตอบ สีที่ต่างกันใช้บอกชนิดของ protocol เช่น สีม่วงคือ HPPT สีเขียวคือ TCP สีส้มหรือน้ำตาลคือ UDP

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP traffic as the clients communicate with the server.







PDU Information at Device: HTTP Client

OSI Model | Inbound PDU Details

PDU Formats

TTL:128	PRO:0x06	CHKSUM
SRC IP:192.168.1.254		
DST IP:192.168.1.1		
DATA (VARIABLE LENGTH)		

TCP

0 4 8 16 24 Bits

SOURCE PORT:80			DESTINATION PORT:1026
SEQUENCE NUMBER:1			
ACKNOWLEDGEMENT NUMBER:103			
OF FS	RE SE	FLAGS:0b00 011000	WINDOW:16384
CHECKSUM:0x0000			URGENT POINTER:0x0000
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

HTTP RESPONSE

0 4 8 16 Bytes

HTTP Data:Connection: close Content-Length: 131

Why did it take so long for the HTTP PDU to appear? ตอบ เพราะคอมต้องทำ ARP เพื่อหา MAC Address ก่อน แล้วก็ต้องทำ TCP Handshake (3 รอบ) ให้เสร็จก่อน ถึงจะเริ่มส่งข้อมูล HTTP ได้ มันเลยช้าหน่อย

What is the section labeled?

ตอบ TCP

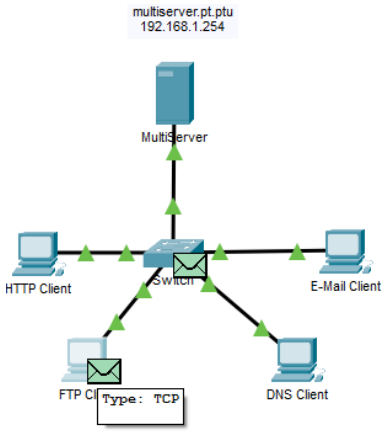
Are these communications considered to be reliable? ตอบ ใช่ (Reliable) เพราะใช้ TCP มันมีการยืนยันการรับส่งข้อมูล

Which TCP flags are set in this PDU? ตอบ SYN (เพราะเพิ่งเริ่มขอเชื่อมต่อ)

How are the port and sequence numbers different than before? ตอบ เลข Port สลับฝั่งกัน (Source กลายเป็น Dest, Dest กลายเป็น Source) ส่วนเลข Ack จะเพิ่มขึ้น 1 เพื่อยืนยันว่าได้รับข้อมูลแล้ว

What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs? ตอบ รอบนี้ Flag เป็น ACK (ยืนยันกลับ) เลข Port สลับกลับมาเหมือนตอนแรกสุด และเลข Sequence ก็ขยับตาม Step การส่ง

Step 2: Examine FTP traffic as the clients communicate with the server.



PDU Information at Device: FTP Client

OSI Model | Outbound PDU Details

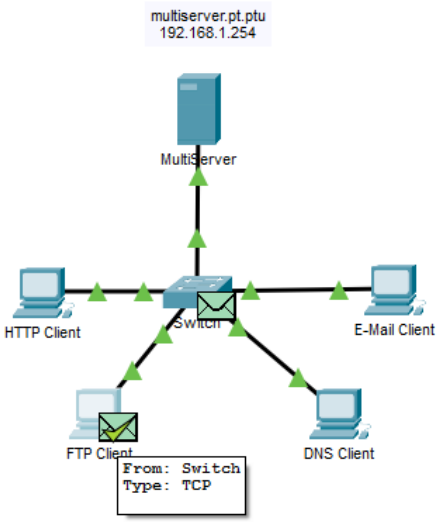
PDU Formats

TTL:128	PRO:0x06	CHKSUM
SRC IP:192.168.1.2		
DST IP:192.168.1.254		
DATA (VARIABLE LENGTH)		

TCP

0 4 8 16 24 Bits

SOURCE PORT:1026			DESTINATION PORT:21
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OF FS	RE SE	FLAGS:0b00 000010	WINDOW:65535
CHECKSUM:0x0000			URGENT POINTER:0x0000
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0



PDU Information at Device: FTP Client

OSI Model | Inbound PDU Details | Outbound PDU Details

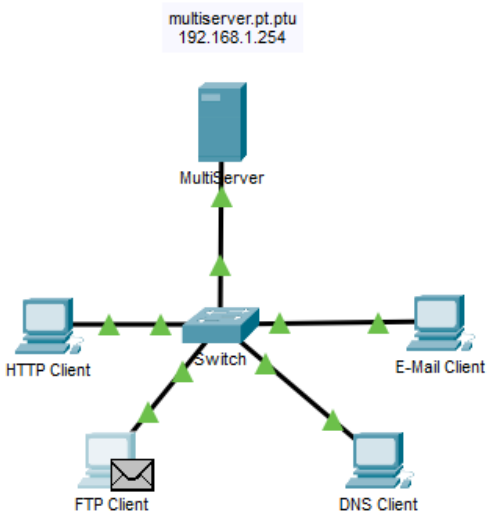
PDU Formats

VER:4	IHL:5	DSCP:0x00	TL:44	
ID:0x0010		F	FRAG OFFSET:0x000	
TTL:128	PRO:0x06	CHKSUM		
SRC IP:192.168.1.254				
DST IP:192.168.1.2				
DATA (VARIABLE LENGTH)				

TCP

0 4 8 16 24 Bits

SOURCE PORT:21		DESTINATION PORT:1026	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:1			
OF FS	RE SE	FLAGS:0b00 010010	WINDOW:16384
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0



PDU Information at Device: FTP Client

OSI Model | Inbound PDU Details

PDU Formats

SRC IP:192.168.1.254
DST IP:192.168.1.2
DATA (VARIABLE LENGTH)

TCP

0 4 8 16 24 Bits

SOURCE PORT:21	DESTINATION PORT:1026
SEQUENCE NUMBER:1	
ACKNOWLEDGEMENT NUMBER:1	
OF RE FLAGS:0b00 011000 FS SE	WINDOW:16384
CHECKSUM:0x0000	URGENT POINTER:0x0000
OPTION	
DATA (VARIABLE LENGTH)	PADDING: 0

FTP Response

0 4 8 16 Bytes

Code:220
Message:Welcome to PT Ftp server

Are these communications considered to be reliable? ตอบ ใช่ เพราะ FTP ก็ใช้ TCP เหมือนกัน

What is the value in the flag field? ตอบ SYN

How are the port and sequence numbers different than before? ตอบ Port สลับฝั่งกัน และเลข Ack เพิ่มขึ้นเพื่อยืนยัน

How are the port and sequence numbers different from the previous results? ตอบ Port สลับกลับมาเหมือนตอนส่งครั้งแรก ส่วนเลข Sequence/Ack ก็เปลี่ยนไปตามลำดับ

What is the message from the server? ตอบ Welcome to PT Ftp server (มันคือข้อความต้อนรับจากเซิร์ฟเวอร์)

Step 3: Examine DNS traffic as the clients communicate with the server.

PDU Information at Device: DNS Client

OSI Model | Outbound PDU Details

At Device: DNS Client
Source: DNS Client
Destination: 192.168.1.254

In Layers	Out Layers
Layer7	Layer 7: DNS
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: UDP Src Port: 1026, Dst Port: 53
Layer3	Layer 3: IP Header Src. IP: 192.168.1.3, Dest. IP: 192.168.1.25
Layer2	Layer 2: Ethernet II Header 000B.BE63.D2C3 >> 0001.96A9.401D
Layer1	Layer 1: Port(s): FastEthernet0

1. The DNS client sends an A DNS query to the DNS server.

PDU Information at Device: DNS Client

OSI Model | Outbound PDU Details

PDU Formats

DATA (VARIABLE LENGTH)

UDP

0 16 Bits

SOURCE PORT:1026	DESTINATION PORT:53
LENGTH:0x002a	CHECKSUM:0
DATA (VARIABLE LENGTH)	

DNS Header

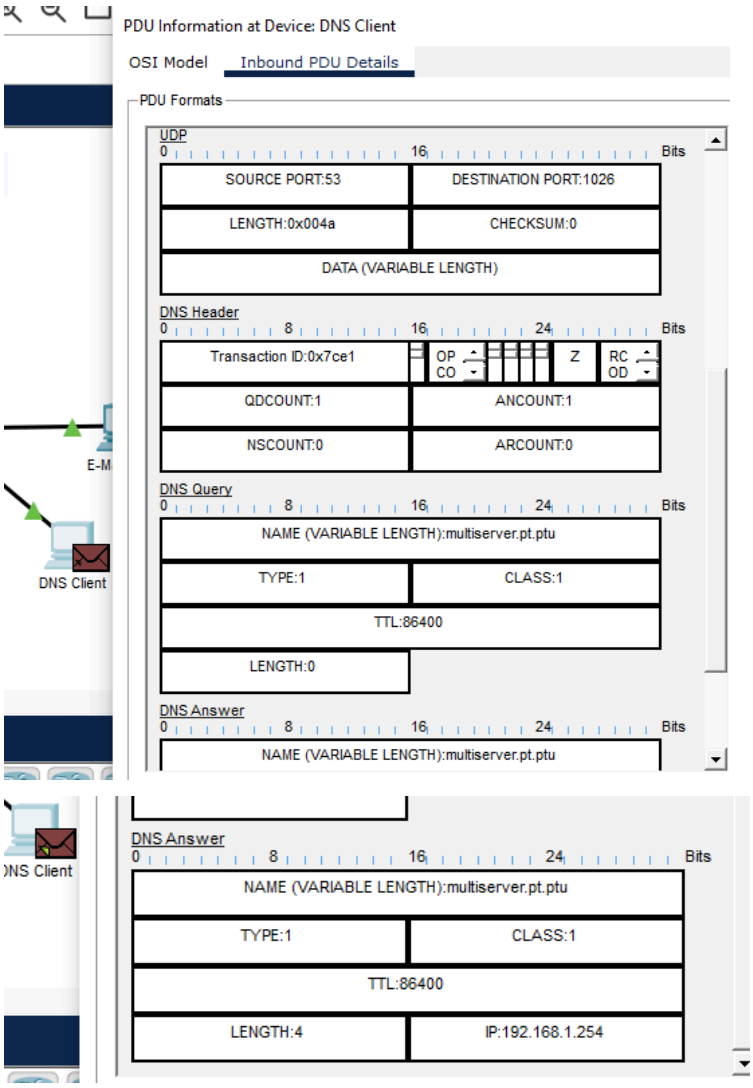
0 8 16 24 Bits

Transaction ID:0x7ce1	OP CO	Z	RC OD
QDCOUNT:1	ANCOUNT:0		
NSCOUNT:0	ARCOUNT:0		

DNS Query

0 8 16 24 Bits

NAME (VARIABLE LENGTH):multiserver.pt.ptu	
TYPE:1	CLASS:1
TTL:86400	
LENGTH:0	



What is the Layer 4 protocol? ตอบ UDP

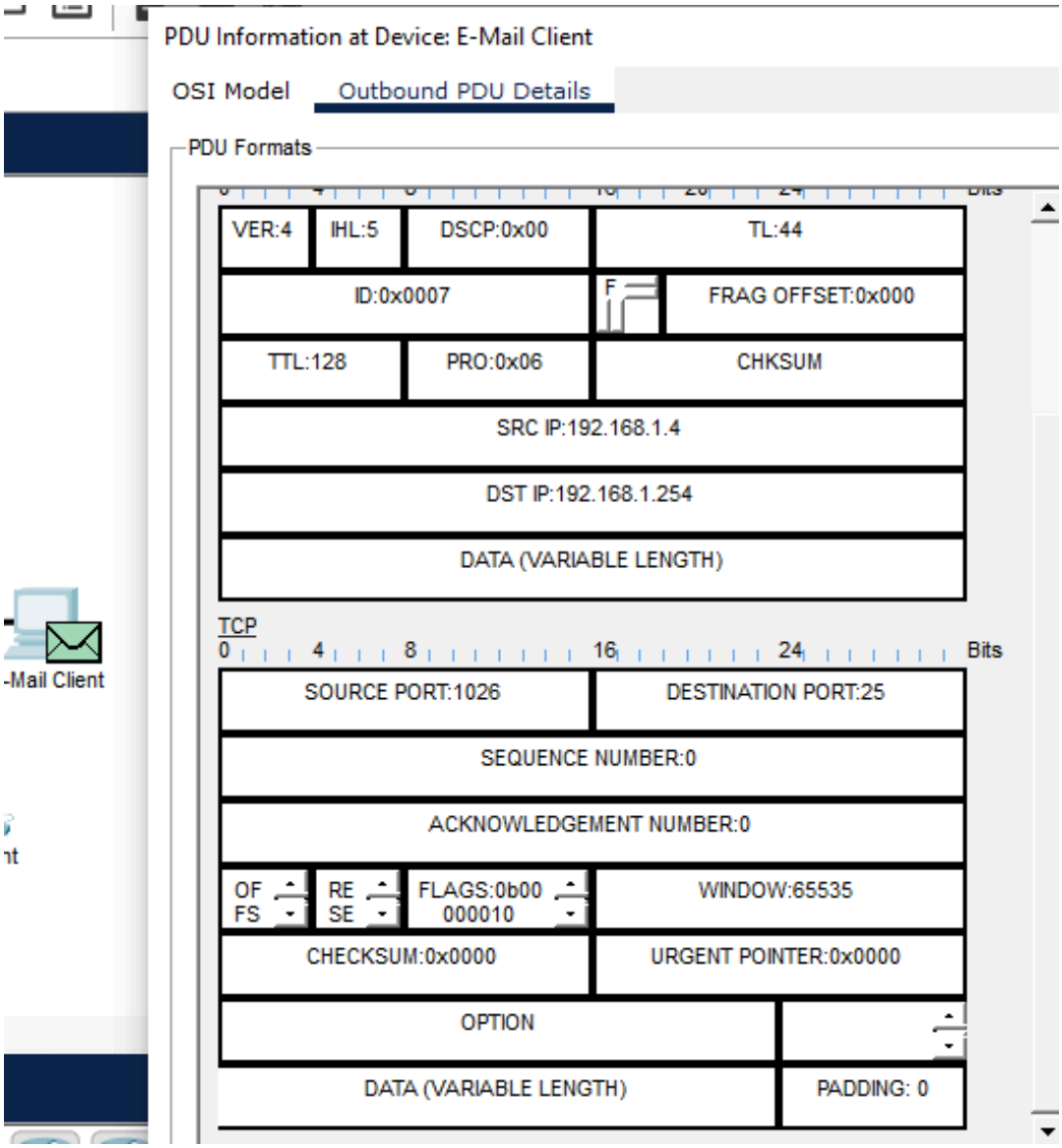
Are these communications considered to be reliable? ตอบ ไม่) เพราะ UDP ส่งแล้วส่งเลย ไม่มีการเช็คว่าถึงไหม

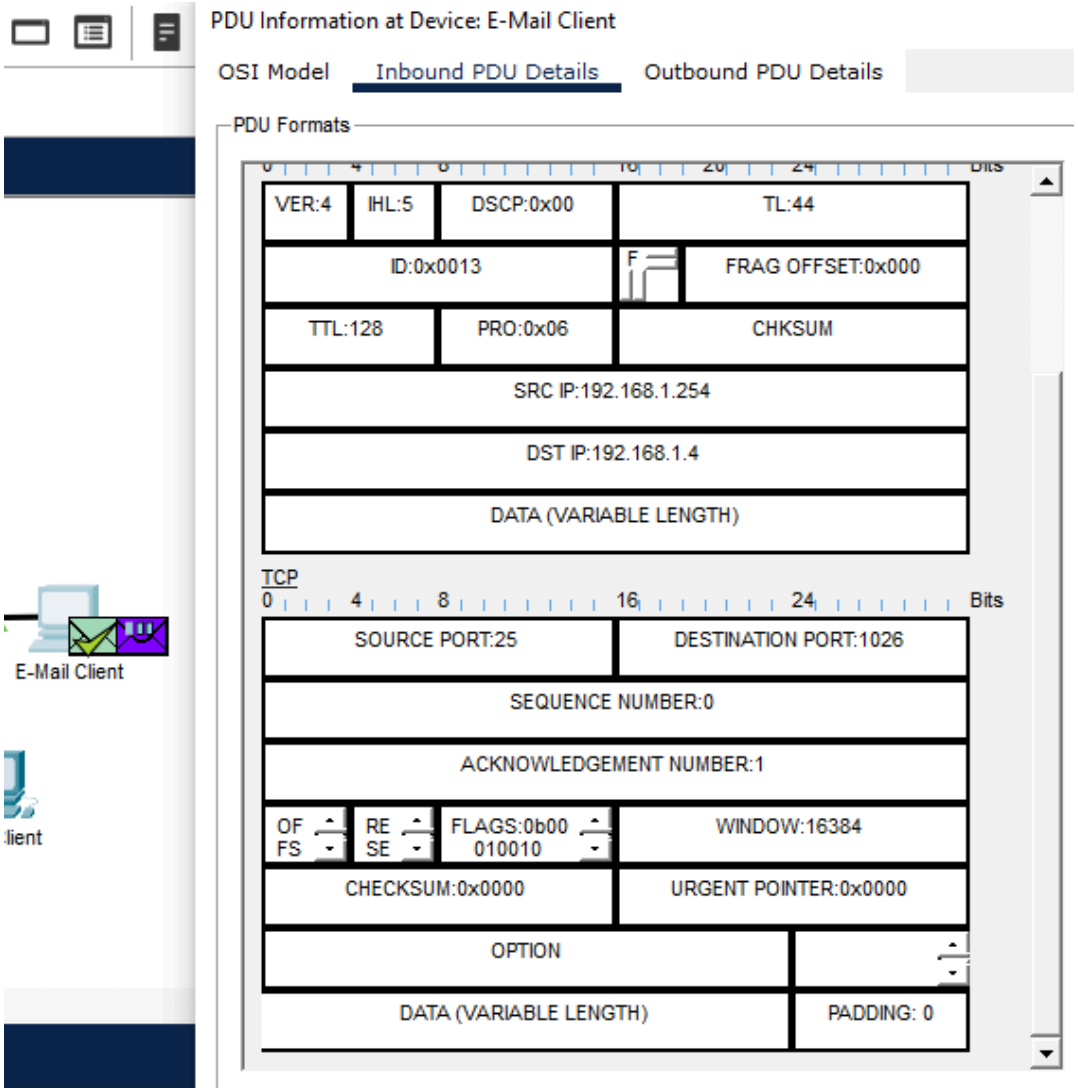
Why are there no sequence and acknowledgement numbers? ตอบ เพราะ UDP ไม่เน้นความชัวร์ ไม่ต้องเรียงลำดับข้อมูล เลยไม่ต้องมีเลขพวกนี้

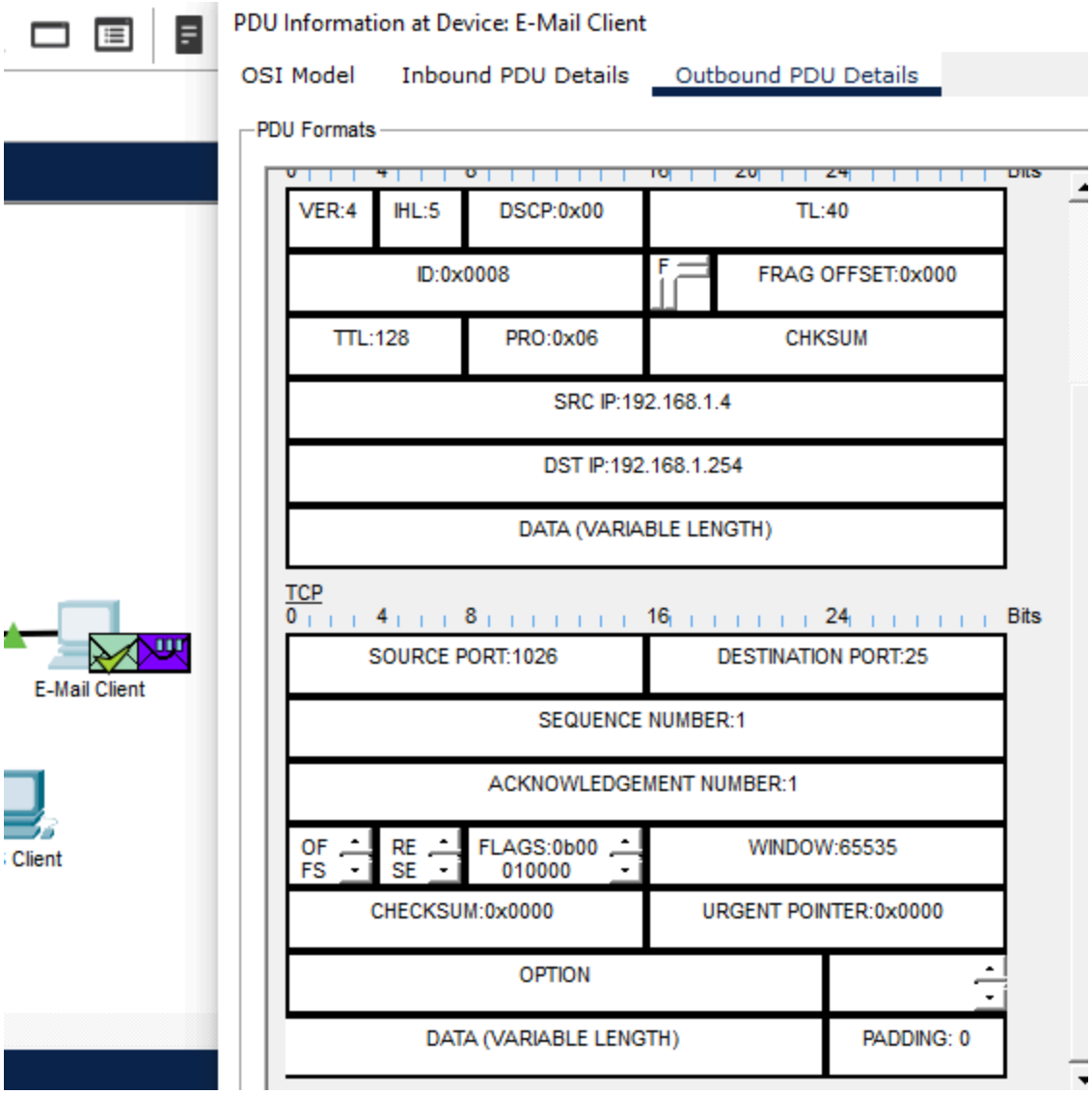
How are the port and sequence numbers different than before? ตอบ Port สลับฝั่งกัน (Source <-> Dest) ไม่มี Seq number ให้เปลี่ยน

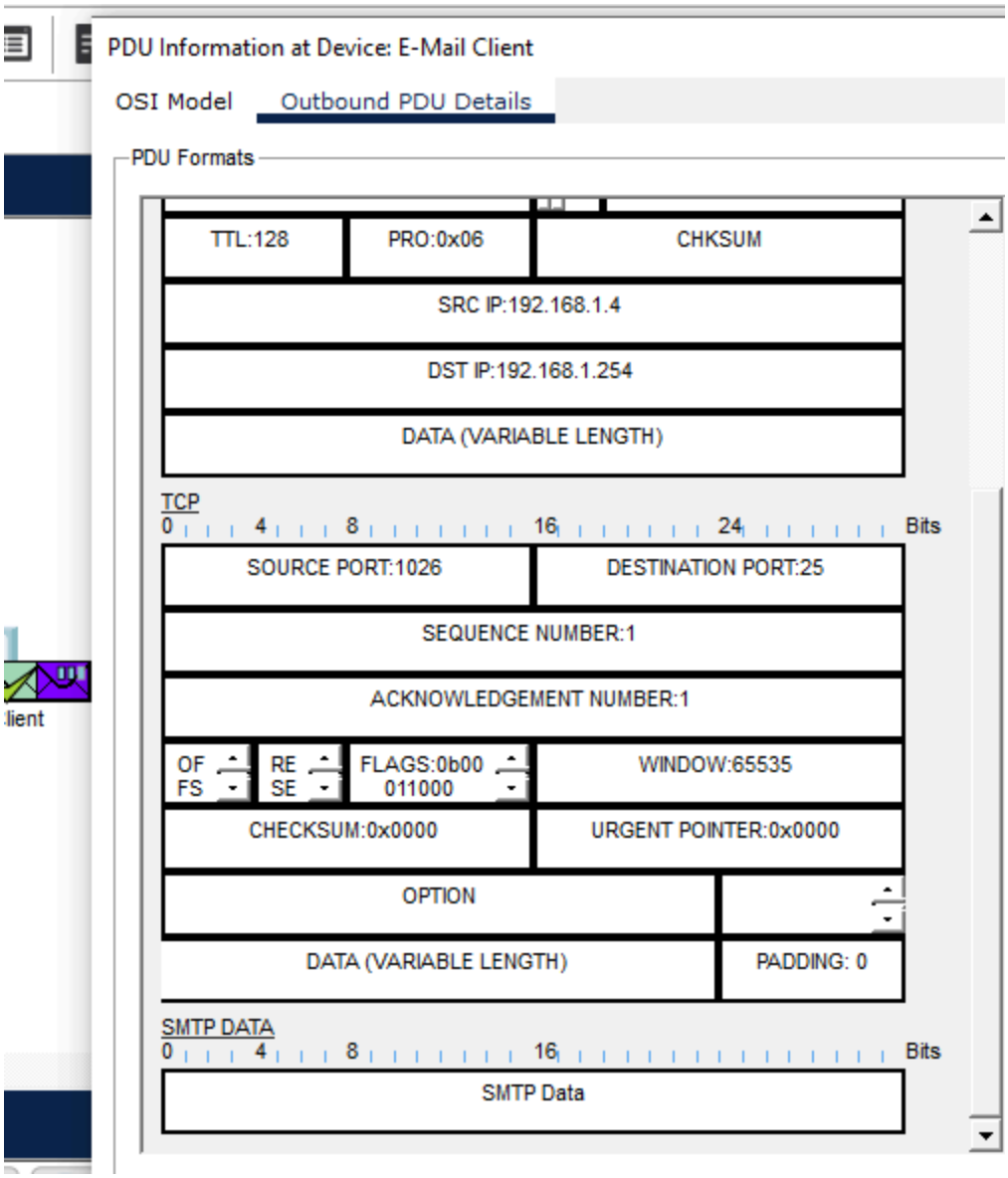
What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?ตอบ ส่วนสุดท้ายเรียกว่า DNS Answer Address / IP ที่ได้คือ 192.168.1.254

Step 4: Examine email traffic as the clients communicate with the server.


PDU Information at Device: E-Mail Client
OSI Model
Outbound PDU Details
PDU Formats
VER:4
IHL:5
DSCP:0x00
TL:44
ID:0x0007
F
FRAG OFFSET:0x000
TTL:128
PRO:0x06
CHKSUM
SRC IP:192.168.1.4
DST IP:192.168.1.254
DATA (VARIABLE LENGTH)
TCP
0
4
8
16
24
Bits
SOURCE PORT:1026
DESTINATION PORT:25
SEQUENCE NUMBER:0
ACKNOWLEDGEMENT NUMBER:0
OF
FS
RE
SE
FLAGS:0b00
000010
WINDOW:65535
CHECKSUM:0x0000
URGENT POINTER:0x0000
OPTION
DATA (VARIABLE LENGTH)
PADDING: 0
-Mail Client


PDU Information at Device: E-Mail Client
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
VER:4
IHL:5
DSCP:0x00
TL:44
ID:0x0013
F
FRAG OFFSET:0x000
TTL:128
PRO:0x06
CHKSUM
SRC IP:192.168.1.254
DST IP:192.168.1.4
DATA (VARIABLE LENGTH)
TCP
0 4 8 16 24 Bits
SOURCE PORT:25
DESTINATION PORT:1026
SEQUENCE NUMBER:0
ACKNOWLEDGEMENT NUMBER:1
OF
FS
RE
SE
FLAGS:0b00
010010
WINDOW:16384
CHECKSUM:0x0000
URGENT POINTER:0x0000
OPTION
DATA (VARIABLE LENGTH)
PADDING: 0
E-Mail Client


PDU Information at Device: E-Mail Client
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
VER:4
IHL:5
DSCP:0x00
TL:40
ID:0x0008
F
FRAG OFFSET:0x000
TTL:128
PRO:0x06
CHKSUM
SRC IP:192.168.1.4
DST IP:192.168.1.254
DATA (VARIABLE LENGTH)
TCP
0
4
8
16
24
Bits
SOURCE PORT:1026
DESTINATION PORT:25
SEQUENCE NUMBER:1
ACKNOWLEDGEMENT NUMBER:1
OF
FS
RE
SE
FLAGS:0b00
010000
WINDOW:65535
CHECKSUM:0x0000
URGENT POINTER:0x0000
OPTION
DATA (VARIABLE LENGTH)
PADDING: 0
E-Mail Client
Client



What transport layer protocol does email traffic use? ตอบ TCP

Are these communications considered to be reliable? ตอบ ใช่ (Reliable)

What is the flag field value? ตอบ SYN

How are the port and sequence numbers different than before? ตอบ Port สลับฝั่งกัน, Ack เพิ่มขึ้น 1

How are the port and sequence numbers different from the previous two results? ตอบ Port สลับกลับมาเหมือนขาไป, Sequence/Ack เปลี่ยนไปตามสเต็ป

How are the port and sequence numbers different from the previous two PDUs? ตอบ เหมือนเดิม คือ Port สลับไป สลับมาตามฝั่งที่ส่ง และเลข Seq/Ack ขยับไปเรื่อยๆ

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

ตอบ Port 25 คือ SMTP (ใช้ส่งเมล), Port 110 คือ POP3 (ใช้รับเมล)